Lecture Notes on Thermodynamic Computing

Chapter 2: kT-RAM and Thermal Memory Technologies

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1 2.1 Introduction to kT-RAM

• What is kT-RAM?

Definition 1.1 (kT-RAM)**.** kT-RAM (thermal-kinetic Random Access Memory) is a memory technology that stores information using energy states at the thermal energy scale $k_B T$, operating through controlled thermal fluctuations.

Key characteristics:

- Energy per bit: $\sim k_B T$ (theoretical minimum)
- Ultra-low voltage operation: < 100 mV
- Probabilistic bit storage and retrieval
- Temperature-dependent operation

• Energy Comparison

Memory Type	Energy per Bit	Ratio to $k_B T$
DRAM	10^{-12} J	10^6
SRAM	10^{-13} J	10^5
Flash	10^{-12} J	10^6
kT-RAM	4×10^{-21} J	1

Table 1: Memory energy comparison at 300K

2 2.2 Physical Principles of kT-RAM

• Thermal Energy Storage

Information stored in discrete energy levels separated by $\sim k_B T$:

$$\Delta E = E_1 - E_0 \approx k_B T \tag{1}$$

Bit error probability:

$$P_{error} = \frac{1}{1 + \exp(\Delta E / k_B T)} \tag{2}$$

For reliable storage ($P_{error} < 10^{-6}$):

$$\Delta E > 14 k_B T \tag{3}$$

- **Capacitive Energy Storage**

Minimum capacitance for thermal storage:

$$C_{min} = \frac{e^2}{k_B T} \approx 6 \text{ aF at 300K} \tag{4}$$

Thermal noise voltage:

$$\langle V_n^2 \rangle = \frac{k_B T}{C} \tag{5}$$

Signal voltage for reliable detection:

$$V_{signal} > 4\sqrt{\frac{k_B T}{C}} \tag{6}$$

- **Retention Time**

Thermal decay constant:

$$\tau = RC = \frac{C \cdot k_B T}{q \cdot I_{leak}} \tag{7}$$

For practical retention ($\tau > 1$ ms):

$$I_{leak} < \frac{C \cdot k_B T}{q \cdot 10^{-3}} \approx 100 \text{ zA} \tag{8}$$

3 2.3 kT-RAM Cell Design

- **Single Transistor kT-Cell**

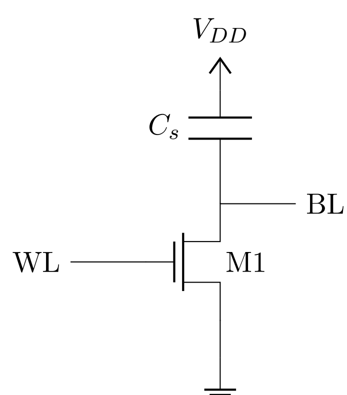

Figure 1: Basic kT-RAM cell

Storage capacitance:

$$C_s = C_{gate} + C_{junction} \approx 10 \text{ aF} \tag{9}$$

- **Write Operation**

Charge injection during write:

$$Q_{stored} = C_s \cdot V_{data} \tag{10}$$

Minimum write voltage:

$$V_{write} > 4\sqrt{\frac{k_B T}{C_s}} \approx 80 \text{ mV at 300K} \tag{11}$$

Write energy:

$$E_{write} = \frac{1}{2} C_s V_{write}^2 \approx 4 k_B T \tag{12}$$

- **Read Operation**

Sense amplifier for small signal detection:

$$V_{signal} = \frac{Q_{stored}}{C_s + C_{parasitic}} \tag{13}$$

Minimum detectable signal:

$$V_{min} = 4\sqrt{\frac{k_B T}{C_{total}}} \tag{14}$$

Read energy (sensing):

$$E_{read} = \frac{1}{2} C_{total} V_{signal}^2 \approx 2 k_B T \tag{15}$$

4 2.4 Thermal Fluctuation Effects

- **Charge Fluctuations**

RMS charge fluctuation:

$$\sigma_Q = \sqrt{k_B T C_s} \approx 0.2 \text{ electrons at 300K} \tag{16}$$

Signal-to-noise ratio:

$$SNR = \frac{Q_{signal}^2}{\sigma_Q^2} = \frac{Q_{signal}^2}{k_B T C_s} \tag{17}$$

For $SNR > 100$ (20 dB):

$$Q_{signal} > 10\sqrt{k_B T C_s} \approx 2 \text{ electrons} \tag{18}$$

- **Temperature Dependence**

Retention time vs temperature:

$$\tau(T) = \tau_0 \exp\left(\frac{E_a}{k_B T}\right) \tag{19}$$

Where E_a is the activation energy for leakage.
Optimal operating temperature:

$$T_{opt} = \frac{E_a}{k_B \ln(f \cdot \tau_{required})} \tag{20}$$

5 2.5 Sense Amplifier Design

- **Thermal Noise Limited Sensing**

Minimum input-referred noise:

$$V_{n,in}^2 = \frac{k_B T}{C_{input}} \tag{21}$$

Transconductance amplifier:

$$A_v = g_m R_{load} \tag{22}$$

Noise figure:

$$NF = 1 + \frac{\gamma g_{m,load}}{g_{m,input}} \tag{23}$$

where $\gamma \approx 2/3$ for long-channel MOSFETs.

- **Regenerative Sense Amplifier**

Positive feedback for signal amplification:

$$\frac{dV}{dt} = \frac{V \cdot (g_{m1} - g_{m2})}{C_{load}} \tag{24}$$

Minimum sensing time:

$$t_{sense} = \frac{C_{load}}{g_m} \ln\left(\frac{V_{DD}}{V_{signal}}\right) \tag{25}$$

Energy per sense operation:

$$E_{sense} = I_{bias} \cdot V_{DD} \cdot t_{sense} \tag{26}$$

6 2.6 Process Variations and Mismatch

- **Threshold Voltage Variations**

Standard deviation of V_{th}:

$$\sigma_{V_{th}} = \frac{A_{V_{th}}}{\sqrt{W \cdot L}} \tag{27}$$

Typical values: $A_{V_{th}} = 5 - 20$ mV·m
Current mismatch in subthreshold:

$$\frac{\sigma_{I_d}}{I_d} = \frac{\sigma_{V_{th}}}{nV_T} \tag{28}$$

- **Capacitance Matching**

Capacitance mismatch:

$$\frac{\sigma_C}{C} = \frac{A_C}{\sqrt{Area}} \tag{29}$$

Impact on sensing:

$$\sigma_{V_{signal}} = V_{signal} \cdot \frac{\sigma_C}{C} \tag{30}$$

Design rule: Keep mismatch $< 10\%$ of thermal noise.

7 2.7 Memory Array Architecture

- **Array Organization**

Bitline capacitance:

$$C_{BL} = n_{rows} \cdot C_{cell} + C_{wire} \tag{31}$$

Signal attenuation:

$$V_{BL} = V_{cell} \cdot \frac{C_{cell}}{C_{cell} + C_{BL}} \tag{32}$$

Optimal array size:

$$n_{opt} = \sqrt{\frac{C_{cell}}{C_{wire/row}}} \tag{33}$$

- **Hierarchical Sensing**

Two-stage sensing for large arrays:

- Local sense amplifiers: 64-128 cells
- Global sense amplifiers: final decision

Energy scaling:

$$E_{total} = E_{local} \cdot \sqrt{N} + E_{global} \tag{34}$$

8 2.8 Advanced kT-RAM Concepts

- **Multi-Level kT-Cells**

Store n bits per cell using 2^n voltage levels:

$$V_k = k \cdot \frac{V_{ref}}{2^n - 1}, \quad k = 0, 1, ..., 2^n - 1 \tag{35}$$

Level spacing for reliability:

$$\Delta V > 4\sqrt{\frac{k_B T}{C_s}} \tag{36}$$

Maximum levels:

$$n_{max} = \log_2 \left(\frac{V_{DD}}{4\sqrt{k_B T / C_s}} \right) \tag{37}$$

- **Refresh-Free Operation**

Using thermal equilibrium for data retention:

$$P(state) = \frac{\exp(-E_{state}/k_B T)}{Z} \tag{38}$$

Partition function:

$$Z = \sum_{states} \exp(-E_{state}/k_B T) \tag{39}$$

Self-correcting through thermal fluctuations.

- **Temperature-Compensated Design**

Voltage scaling with temperature:

$$V_{ref}(T) = V_0\sqrt{\frac{T}{T_0}} \tag{40}$$

Maintains constant V_{ref}/V_T ratio.

9 2.9 Applications and Limitations

- **Suitable Applications**
  - IoT sensor nodes with ultra-low power requirements
  - Probabilistic computing and machine learning
  - Approximate computing applications
  - Cache memory for thermodynamic processors

- **Limitations**
  - Temperature sensitivity
  - Process variation sensitivity
  - Limited operating speed
  - Complex peripheral circuits

- **Performance Projections**

Parameter	Current DRAM	kT-RAM (projected)	Improvement
Energy/bit	1 pJ	10 aJ	100×
Retention	64 ms	1 ms	0.015×
Access time	10 ns	1 s	0.01×
Density	8 Gb/cm²	1 Tb/cm²	125×

Table 2: kT-RAM vs conventional memory

10 Examples

Example 10.1 (kT-Cell Energy Calculation)**.** Design kT-RAM cell for $E_{write} = 5k_BT$ at 300K.

Given:

- $T = 300$ K
- $k_B = 1.38 \times 10^{-23}$ J/K
- Target energy: $5k_BT$

Solution:
Target energy:

$$E_{target} = 5 \times 1.38 \times 10^{-23} \times 300 \tag{41}$$
$$= 2.07 \times 10^{-20} \text{ J} \tag{42}$$

From $E = \frac{1}{2}CV^2$, with $V = 4V_T = 104$ mV:

$$C = \frac{2E}{V^2} = \frac{2 \times 2.07 \times 10^{-20}}{(0.104)^2} \tag{43}$$
$$= 3.82 \times 10^{-18} \text{ F} = 3.82 \text{ aF} \tag{44}$$

Cell dimensions for gate capacitance:

$$C = \epsilon_{ox} \frac{W \cdot L}{t_{ox}} \tag{45}$$
$$3.82 \times 10^{-18} = 3.9 \times 8.85 \times 10^{-12} \frac{W \cdot L}{2 \times 10^{-9}} \tag{46}$$

Required area: $W \cdot L = 22$ nm²

Example 10.2 (Retention Time Analysis)**.** Calculate retention time for kT-cell with $C = 5$ aF, $I_{leak} = 1$ zA.
Solution:
Thermal voltage at 300K:

$$V_T = \frac{k_B T}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} \tag{47}$$
$$= 25.9 \text{ mV} \tag{48}$$

RC time constant:

$$\tau = \frac{C \cdot V_T}{I_{leak}} \tag{49}$$
$$= \frac{5 \times 10^{-18} \times 25.9 \times 10^{-3}}{1 \times 10^{-21}} \tag{50}$$
$$= 129.5 \text{ seconds} \tag{51}$$

Data retention for 99

$$t_{99\%} = 4.6\tau = 595 \text{ seconds} \approx 10 \text{ minutes} \tag{52}$$

11 Homework Problems

Assignment 2.1: kT-RAM and Thermal Memory

1. **Energy Scaling Analysis**:
   - Calculate minimum energy for 1-bit storage at 77K, 300K, 400K
   - Design cell capacitance for each temperature

- Find optimal operating voltage

2. **Retention Time Design**:
   - Target retention: 1 second at 300K
   - Maximum leakage current allowed
   - Cell area constraints for 5 aF capacitance

3. **Sense Amplifier Design**:
   - Design for 10 mV input signal
   - Calculate required transconductance
   - Find minimum sensing energy

4. **Array Architecture**:
   - Design 1 Kb kT-RAM array
   - Optimize bitline length
   - Calculate total array energy

5. **Process Variation Effects**:
   - V_{th} variation: $\sigma = 50$ mV
   - Impact on cell current mismatch
   - Design margins for reliable operation

6. **Multi-Level Cell Design**:
   - 2-bit per cell kT-RAM
   - Level spacing for 10^{-6} error rate
   - Energy comparison with single-level

7. **Temperature Compensation**:
   - Design reference voltage generator
   - Maintain constant SNR over 0-70°C
   - Power overhead analysis

8. **Comparative Analysis**:
   - kT-RAM vs DRAM energy efficiency
   - Performance trade-offs
   - Application suitability analysis

Next Chapter Preview: Chapter 3 will cover "Stochastic Computing Architectures" including bit-stream processing and noise-resilient circuits.